

Intelligence Processing Unit (IPU) Training

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Host Institution: Prairie View A&M University

Workshop Venue: University of New Haven

August 7, 2025



High Performance
Research Computing
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Agenda

- Introduction to computing systems platform (ACCESS and ACES)
- Introduction to IPU
- Hands on training on IPU

Introduction to ACCESS and ACES



- ACCESS (Advanced Cyberinfrastructure for Computational and Emerging Sciences) is a program funded by the NSF, designed to help researchers and educators with using high-performance computing resources and cyberinfrastructure. Its aim is to make these resources more accessible to researchers. No cost.
- Involves many universities and HPC centers, including:
 - Texas A&M University (TAMU) — part of the broader consortium.
 - San Diego Supercomputer Center (SDSC)
 - Pittsburgh Supercomputing Center (PSC)
 - NCSA (at University of Illinois Urbana-Champaign)
 - TACC (Texas Advanced Computing Center)
 - University of Colorado Boulder
 - and others.
- Texas A&M High Performance Research Computing (HPRC) offers 6 main clusters: Launch, ACES, FASTER, Grace, Portal, Galaxy.

Introduction to ACES

- ACES cluster in HPRC has a variety of HPC resources: Graphcore Intelligent Processing Unit (IPU), NVidia A100/H100 GPUs, Intel Max GPUs, TPU, and FPGA systems. -Comprehensive testbed: Various HPC

ACES: A Dell x86 HPC Cluster

System Name:	ACES
Operating System:	Red Hat Enterprise 8
Total Compute Cores/Nodes:	11,888 cores 130 nodes
Compute Nodes:	110 Intel Sapphire Rapids Nodes 17 Intel Ice Lake Nodes 1 AMD Rome Graphcore Node with 16 Mk2 Colossus GC200 IPUs 1 Intel Ice Lake Graphcore Node with 16 Bow-2000 IPUs 1 Intel Cascade Lake Node with 8 NEC Vector Engine Type 20B-P cards
Composable Components	30 NVIDIA H100 GPUs 4 NVIDIA A30 GPUs 3 Bittware Agilex FPGAs 2 Intel D5005 FPGAs 2 NextSilicon coprocessors 48 Intel Optane SSDs 120 Intel GPU Max 1100
Interconnect:	NVIDIA Mellanox NDR200 InfiniBand (MPI and storage) Liquid PCIe Gen4 Fabrics (composability) Liquid PCIe Gen5 Fabrics (composability)
Peak Performance:	1.6 PFLOPS
Global Disk:	2.6PB (usable) via a DDN Lustre appliance
File System:	Lustre
Batch Facility:	Slurm by SchedMD

- Comprehensive Testbed: Various HPC



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ACES Accelerators

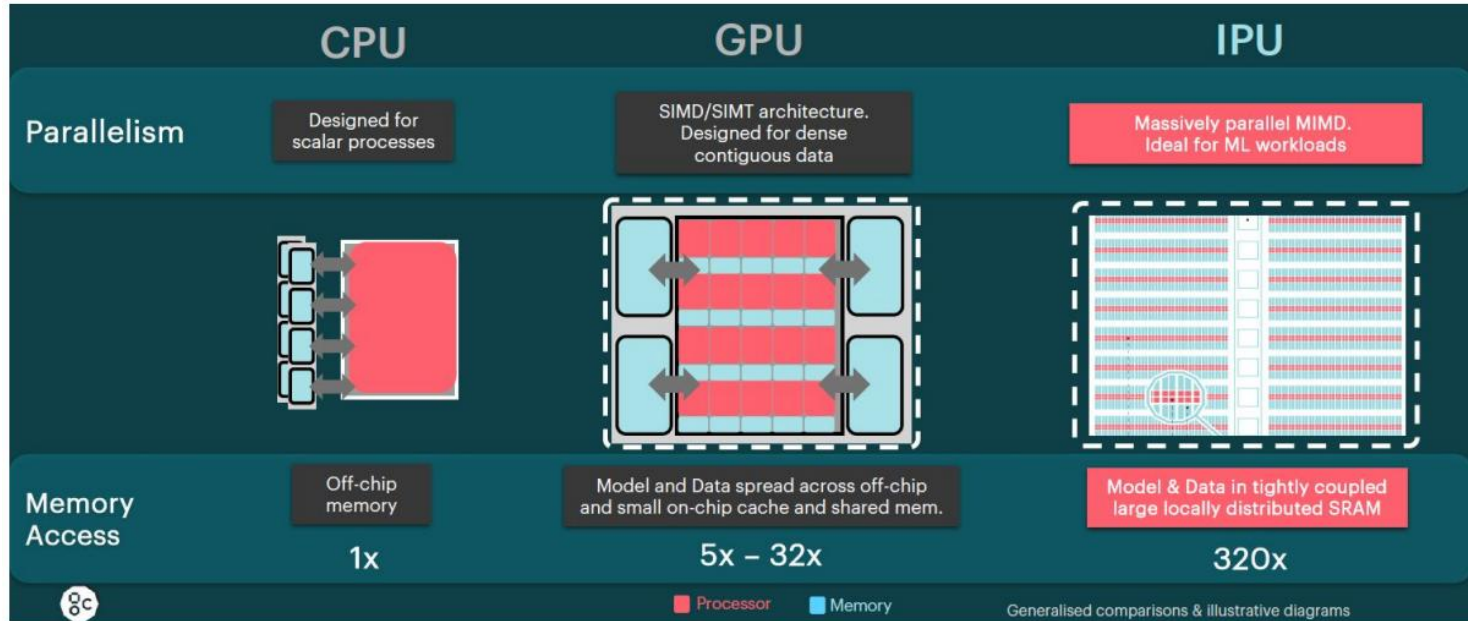
Component	Quantity	Description
Graphcore IPU	32	16 Colossus GC200 IPU, 16 Bow IPU. Each IPU group hosted with a CPU server as a POD16 on a 100 GbE RoCE fabric
<i>FPGAs:</i>		
Intel PAC D5005	2	Accelerator with Intel Stratix 10 GX FPGA and 32 GB DDR4
BittWare IA-840F	3	Accelerator with Agilex AGF027 FPGA and 64 GB of DDR4
NextSilicon Coprocessor	2	Reconfigurable accelerator with an optimizer continuously evaluating application behavior.
NEC Vector Engine	8	Vector computing card (8 cores and HBM2 memory)
Intel Optane SSD	48	18 TB of SSDs addressable as memory w/ MemVerge Memory Machine.
<i>NVIDIA GPUs:</i>		
H100	30	For HPC, DL Training, AI Inference
A30	4	For AI Inference and Mainstream Compute
Intel PVC GPUs	120	Intel GPUs for HPC, DL Training, AI Inference

Refer to our Knowledge Base for more:

<https://hprc.tamu.edu/kb/User-Guides/ACES/Hardware/>

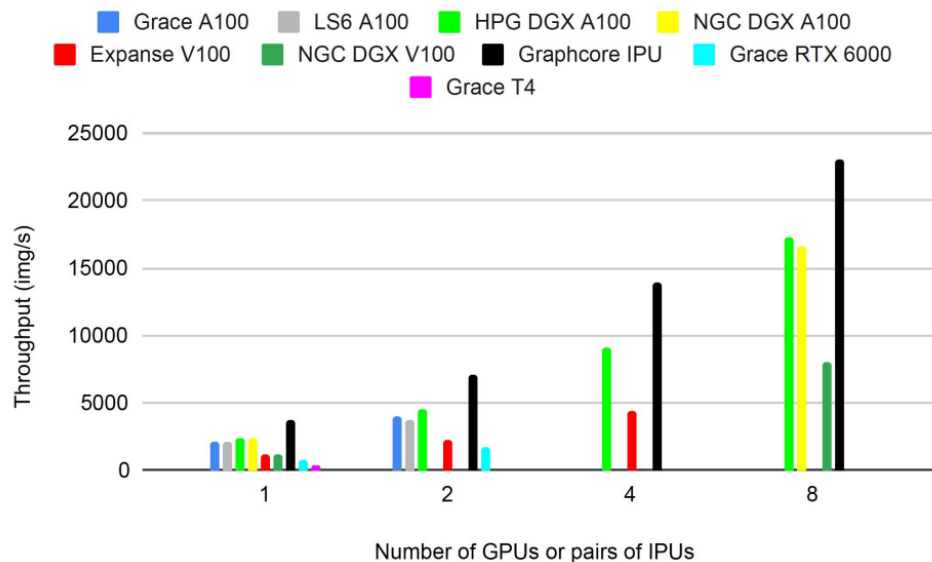
Choosing Between IPUs and GPUs

Introduction to Intelligence Processing Unit (IPU)



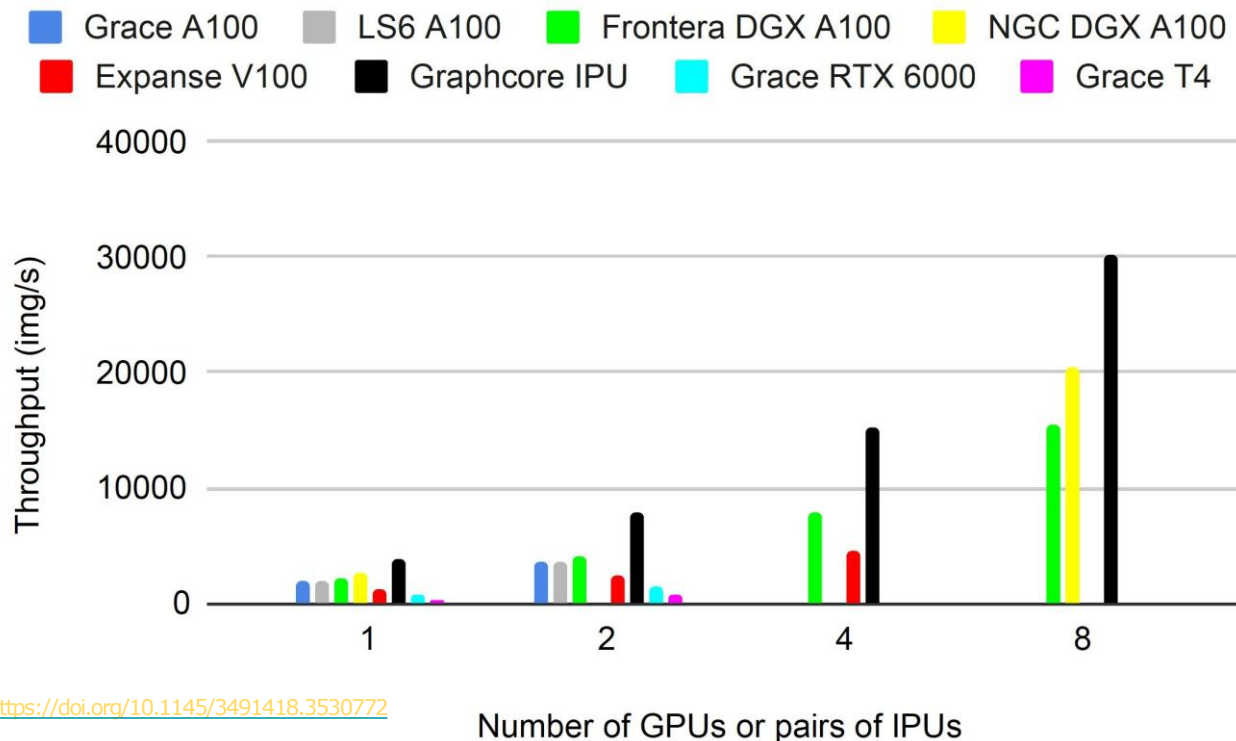
- ❑ **CPU:** General-purpose processor found in most computers. Uses off-chip memory, with low memory access efficiency .
- ❑ **GPU:** Uses SIMD/SIMT architecture for parallel processing of dense data. Data and models are spread across **off-chip and on-chip memory**, which improves computational efficiency.
- ❑ **IPU uses a massively parallel MIMD architecture** and stores both model and data in **on-chip SRAM**. Unlike CPUs and GPUs that rely on off-chip memory. This reduces data movement and offers **efficient, high-speed memory access**, making IPUs ideal for ML.

Pytorch ResNet-50 - GPU vs IPU



- Comparison of GPUs and IPUs on the ResNet 50 model in PyTorch with mixed precision.
- Scaling is similar for GPUs and IPUs, with similar magnitude per-device.
- NVIDIA A100 series GPUs are indeed powerful, when compared to the Graphcore IPU, they still fall short by a considerable margin.

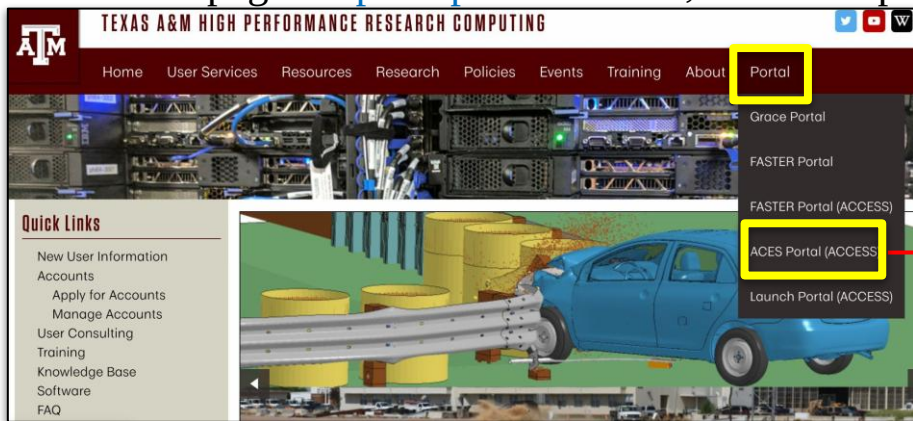
TensorFlow ResNet-50 - GPU vs IPU



Abhinand S. Nasari, et al. <https://doi.org/10.1145/3491418.3530772>

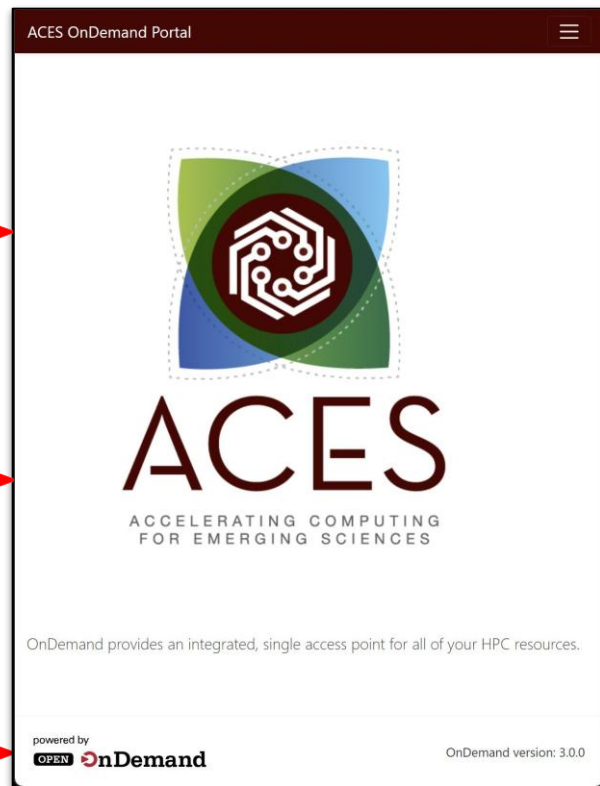
Log into ACES Using the HPRC Portal

HPRC webpage: <https://hprc.tamu.edu/>, Portal dropdown menu



ACES Portal adopts Open OnDemand as its web interface.

Open OnDemand (OOD) is a web-based graphical interface framework designed for HPC users.



Accessing ACES via the Portal (ACCESS)

Log-in using your ACCESS credentials.

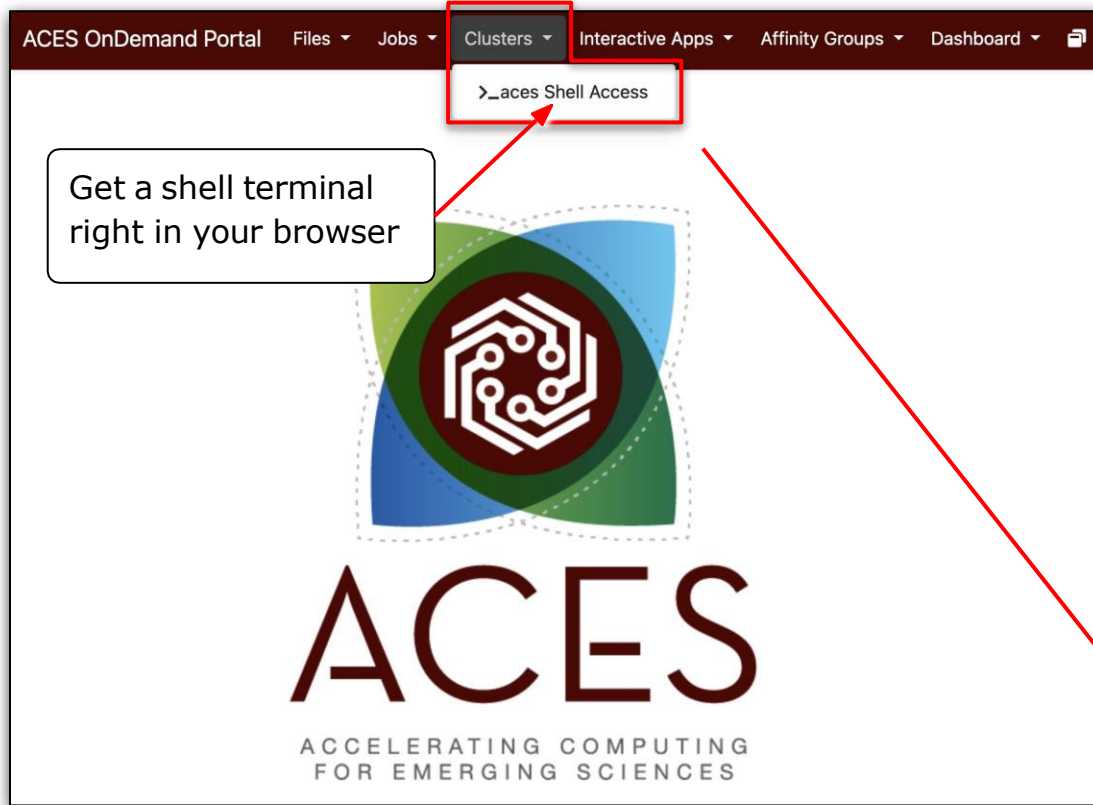
The screenshot shows the ACCESS portal login page. At the top, there is a teal banner with the ACCESS logo on the left and 'Powered By CILogon' on the right. Below the banner is a white box titled 'Consent to Attribute Release' with a dropdown arrow. Inside this box, it says 'TAMU FASTER ACCESS (OO)' requests access to the following information. If you do not approve this request, do not proceed. Below this are four bullet points: 'Your CILogon user identifier', 'Your name', 'Your email address', and 'Your username and affiliation from your identity provider'. Below the consent box is a teal box titled 'Select an Identity Provider'. Inside this box, there is a dropdown menu showing 'ACCESS CI (XSEDE)' with a question mark icon. Below the dropdown is a checkbox labeled 'Remember this selection' and a 'Log On' button. At the bottom of the box, it says 'By selecting "Log On", you agree to the privacy policy.' Below the 'Select an Identity Provider' box is a teal footer with text: 'For questions about this site, please see [link] or send email to [link] using the CILogon Service. See acknowledgements of support for this site.'

This is a close-up of the 'Select an Identity Provider' dropdown menu. It shows a teal header with the text 'Select an Identity Provider'. Below the header is a white box with a dropdown menu showing 'ACCESS CI (XSEDE)' with a question mark icon.

Select the Identity Provider appropriate for your account.

The screenshot shows the ACCESS portal login page. At the top, there is a teal banner with the ACCESS logo on the left and 'Powered By CILogon' on the right. Below the banner is a white box titled 'Login to CILogon'. Inside this box, there are two input fields: 'ACCESS Username' and 'ACCESS Password'. Below the password field is a checkbox labeled 'Don't Remember Login' and a 'Login' button. To the right of the login box is the CILogon logo and the text 'CILogon facilitates secure access to CyberInfrastructure (CI)'. Below the logo are four links: 'If you had an XSEDE account, please enter your XSEDE username and password for ACCESS login', 'Register for an ACCESS Account', 'Forgot your password?', and 'Need Help?'. At the bottom of the page is a teal footer with the text 'Click Here for Assistance'.

Shell Access via the Portal



```
Host: login.aces Themes: Default
*****
This computer system and the data herein are available only for authorized
purposes by authorized users. Use for any other purpose is prohibited and may
result in disciplinary actions or criminal prosecution against the user. Usage
may be subject to security testing and monitoring. There is no expectation of
privacy on this system except as otherwise provided by applicable privacy laws.
Refer to University SAP 29.01.03.M0.02 Acceptable Use for more information.
*****
Last login: Wed Mar 13 09:55:42 2024 from 10.71.1.6

=====
Texas A&M University High Performance Research Computing
=====
Website: https://hprc.tamu.edu
Consulting: help@hprc.tamu.edu (preferred) or (979) 845-0219
ACES Documentation: https://hprc.tamu.edu/kb/User-Guides/ACES
FASTER Documentation: https://hprc.tamu.edu/kb/User-Guides/FASTER
Grace Documentation: https://hprc.tamu.edu/kb/User-Guides/Grace
Terra Documentation: https://hprc.tamu.edu/kb/User-Guides/Terra
YouTube Channel: https://www.youtube.com/texasamhprc
=====

*****
== IMPORTANT POLICY INFORMATION ==
* - Unauthorized use of HPRC resources is prohibited and subject to
* criminal prosecution.
* - Use of HPRC resources in violation of United States export control
* laws and regulations is prohibited. Current HPRC staff members are
* US citizens and legal residents.
* - Sharing HPRC account and password information is in violation of
* Texas State Law. Any shared accounts will be DISABLED.
* - Authorized users must also adhere to ALL policies at:
* https://hprc.tamu.edu/policies/
*****

**** ACES Update, March 7 ****

The pvc queue has been updated with a new set of nodes with 2x, 4x, and 8x PVCs.

!! WARNING: THERE ARE ONLY NIGHTLY BACKUPS OF USER HOME DIRECTORIES. !!

Please restrict usage to 8 CORES across ALL login nodes.
Users found in violation of this policy will be SUSPENDED.

To see these messages again, run the moid command.
Your current disk quotas are:
Disk          Disk Usage    Limit   File Usage    Limit
/home/u.zh108696 5.4G        10.0G    3148         10000
/scratch/user/u.zh108696 439.2G      1.0T    1169787      2000000
Type 'showquota' to view these quotas again.
[u.zh108696@aces-login2 ~]$
```

Shell access via the HPRC Portal

Host: login.aces

Warning: Permanently added 'login.aces,10.71.1.13' (ECDSA) to the list of known hosts.

This computer system and the data herein are available only for authorized purposes by authorized users. Use for any other purpose is prohibited and may result in disciplinary actions or criminal prosecution against the user. Usage may be subject to security testing and monitoring. There is no expectation of privacy on this system except as otherwise provided by applicable privacy laws. Refer to University SAP 29.01.03.M0.02 Acceptable Use for more information.

Last login: Fri Jul 14 16:28:15 2023 from 10.71.1.6

=====

Texas A&M University High Performance Research Computing

=====

Website: <https://hprc.tamu.edu>
Consulting: help@hprc.tamu.edu (preferred) or (979) 845-0219
ACES Documentation: <https://hprc.tamu.edu/kb/User-Guides/ACES>
FASTER Documentation: <https://hprc.tamu.edu/kb/User-Guides/FASTER>
Grace Documentation: <https://hprc.tamu.edu/kb/User-Guides/Grace>
Terra Documentation: <https://hprc.tamu.edu/kb/User-Guides/Terra>
YouTube Channel: <https://www.youtube.com/texasamhprc>
=====

```
*****
*                               === IMPORTANT POLICY INFORMATION ===                               *
* - Unauthorized use of HPRC resources is prohibited and subject to criminal prosecution.          *
* - Use of HPRC resources in violation of United States export control laws and regulations is      *
*   prohibited. Current HPRC staff members are US citizens and legal residents.                  *
* - Sharing HPRC account and password information is in violation of Texas State Law. Any shared    *
*   accounts will be DISABLED.                                                                    *
* - Authorized users must also adhere to ALL policies at:                                          *
*   https://hprc.tamu.edu/policies/                                                                *
*****
```

!! WARNING: THERE ARE ONLY NIGHTLY BACKUPS OF USER HOME DIRECTORIES. !!

Please restrict usage to 8 CORES across ALL login nodes.
Users found in violation of this policy will be SUSPENDED.

To see these messages again, run the `motd` command.

Your current disk quotas are:

Disk	Disk Usage	Limit	File Usage	Limit
/home/u.lz104503	9.0G	10.0G	414	10000
/scratch/user/u.lz104503	27.2G	1.0T	41904	250000

Type 'showquota' to view these quotas again.
[u.lz104503@aces-login3 ~]\$

Training Materials

From the ACES login node, ssh into the poplar2 (BOW Pod16) IPU system

```
ssh poplar2
```

Change to the \$SCRATCH directory- your temporary working directory:

```
cd $SCRATCH
```

Copy the example materials to your SCRATCH directory:

```
git clone https://github.com/graphcore/examples.git
```

Copy the hands-on exercise materials to your SCRATCH directory:

```
git clone https://github.com/happidencel/IPU-Training.git
```

"Use the 'ls' command to check what resources are available in the 'scratch' directory."

Poplar SDK setup

```
# activate the poplar SDK
source /usr/local/bin/source.poplar.sh

mkdir -p tmp

# Set the cache directory for PopTorch
export TF_POPLAR_FLAGS=--executable_cache_path=$SCRATCH/tmp
export POPTORCH_CACHE_DIR=$SCRATCH/tmp
export TORCH_HOME=$SCRATCH/tmp
```

IPU support a variety of deep learning frameworks

- ❑ Open-source deep learning frameworks: Tensorflow, Keras, Pytorch, **Caffe**, **MXNet**, and so on
- ❑ Frameworks are widely used in the research and development of machine learning and artificial intelligence applications.
- ❑ Frameworks provide a comprehensive set of tools and libraries for building and training deep neural networks.



LAB-1

Run a TensorFlow (TF) model on IPU

TF Virtual Environment Setup

- `virtualenv -p python3 venv_tf2` # create virtual python environment
 - `source venv_tf2/bin/activate` # activate environment
- # Install a custom version of TensorFlow optimized for Graphcore IPUs.
- `python -m pip install /opt/gc/poplar/poplar_sdk-ubuntu_20_04-3.3.0+1403-208993bbb7/tensorflow-2.6.3+gc3.3.0+251582+08d96978c7f+intel_skylake512-cp38-cp38-linux_x86_64.whl`

Run a TensorFlow model on IPU

- `cd examples/tutorials/tutorials/tensorflow2/keras/completed_demos/`
- `python completed_demo_ipu.py`

- Deactivate the virtual environment after the model finishes running.

- `deactivate`

Monitor IPU Usage - *gc-monitor*

- 4 partitions
- 16 IPUUs
- Processes

```
watch -n 2 gc-monitor
```

```
"Ctrl + C " to  
interrupt process.
```

Every 2.0s: gc-monitor

poplar2: Thu Jul 6 10:33:31 2023

gc-monitor Partition: p17 [active] has 16 reconfigurable IPUUs									
IPU-M	Serial	IPU-M SW	Server version	ICU FW	Type	ID	IPU#	Routing	
10.5.5.1	0019.0002.8222521	2.6.0	1.11.0	2.5.9	M2000	0	3	DNC	
10.5.5.1	0019.0002.8222521	2.6.0	1.11.0	2.5.9	M2000	1	2	DNC	
10.5.5.1	0019.0001.8222521	2.6.0	1.11.0	2.5.9	M2000	2	1	DNC	
10.5.5.1	0019.0001.8222521	2.6.0	1.11.0	2.5.9	M2000	3	0	DNC	
10.5.5.2	0021.0002.8222521	2.6.0	1.11.0	2.5.9	M2000	4	3	DNC	
10.5.5.2	0021.0002.8222521	2.6.0	1.11.0	2.5.9	M2000	5	2	DNC	
10.5.5.2	0021.0001.8222521	2.6.0	1.11.0	2.5.9	M2000	6	1	DNC	
10.5.5.2	0021.0001.8222521	2.6.0	1.11.0	2.5.9	M2000	7	0	DNC	
10.5.5.3	0013.0002.8222521	2.6.0	1.11.0	2.5.9	M2000	8	3	DNC	
10.5.5.3	0013.0002.8222521	2.6.0	1.11.0	2.5.9	M2000	9	2	DNC	
10.5.5.3	0013.0001.8222521	2.6.0	1.11.0	2.5.9	M2000	10	1	DNC	
10.5.5.3	0013.0001.8222521	2.6.0	1.11.0	2.5.9	M2000	11	0	DNC	
10.5.5.4	0016.0002.8222521	2.6.0	1.11.0	2.5.9	M2000	12	3	DNC	
10.5.5.4	0016.0002.8222521	2.6.0	1.11.0	2.5.9	M2000	13	2	DNC	
10.5.5.4	0016.0001.8222521	2.6.0	1.11.0	2.5.9	M2000	14	1	DNC	
10.5.5.4	0016.0001.8222521	2.6.0	1.11.0	2.5.9	M2000	15	0	DNC	
Attached processes in partition p17				IPU			Board		
PID	Command	Time	User	ID	Clock	Temp	Temp	Power	
902631	python	50s	u.zh108696	0	1500MHz	23.5 C	21.9 C	90.3 W	

LAB-2

Run a PyTorch (PopTorch) model on IPU

PopTorch Virtual Environment Setup

- `cd $SCRATCH`
- `virtualenv -p python3 poptorch_test`
- `source poptorch_test/bin/activate`
- `python -m pip install /opt/gc/poplar/poplar_sdk-ubuntu_20_04-3.3.0+1403-208993bbb7/poptorch-3.3.0+113432_960e9c294b_ubuntu_20_04-cp38-cp38-linux_x86_64.whl` #Install PopTorch, Graphcore's PyTorch interface

Run a PopTorch model on IPU

- `cd examples/tutorials/simple_applications/pytorch/mnist/`
- `python mnist_poptorch.py`

- Deactivate the virtual environment after the model finishes running.

- `deactivate`

Monitor the IPU usage with gc-monitor command

```
watch -n 2 gc-monitor
```

"Ctrl + C " to interrupt gc-monitor process.

- 4 partitions
- 16IPUs
- Processes
- IPU used
- Temperature
- Power

Every 2.0s: gc-monitor poplar2: Thu Jul 6 10:55:55 2023

gc-monitor Partition: p17 [active] has 16 reconfigurable IPU's								
IPU-M	Serial	IPU-M SW	Server version	ICU FW	Type	ID	IPU#	Routing
10.5.5.1	0019.0002.8222521	2.6.0	1.11.0	2.5.9	M2000	0	3	DNC
10.5.5.1	0019.0002.8222521	2.6.0	1.11.0	2.5.9	M2000	1	2	DNC
10.5.5.1	0019.0001.8222521	2.6.0	1.11.0	2.5.9	M2000	2	1	DNC
10.5.5.1	0019.0001.8222521	2.6.0	1.11.0	2.5.9	M2000	3	0	DNC
10.5.5.2	0021.0002.8222521	2.6.0	1.11.0	2.5.9	M2000	4	3	DNC
10.5.5.2	0021.0002.8222521	2.6.0	1.11.0	2.5.9	M2000	5	2	DNC
10.5.5.2	0021.0001.8222521	2.6.0	1.11.0	2.5.9	M2000	6	1	DNC
10.5.5.2	0021.0001.8222521	2.6.0	1.11.0	2.5.9	M2000	7	0	DNC
10.5.5.3	0013.0002.8222521	2.6.0	1.11.0	2.5.9	M2000	8	3	DNC
10.5.5.3	0013.0002.8222521	2.6.0	1.11.0	2.5.9	M2000	9	2	DNC
10.5.5.3	0013.0001.8222521	2.6.0	1.11.0	2.5.9	M2000	10	1	DNC
10.5.5.3	0013.0001.8222521	2.6.0	1.11.0	2.5.9	M2000	11	0	DNC
10.5.5.4	0016.0002.8222521	2.6.0	1.11.0	2.5.9	M2000	12	3	DNC
10.5.5.4	0016.0002.8222521	2.6.0	1.11.0	2.5.9	M2000	13	2	DNC
10.5.5.4	0016.0001.8222521	2.6.0	1.11.0	2.5.9	M2000	14	1	DNC
10.5.5.4	0016.0001.8222521	2.6.0	1.11.0	2.5.9	M2000	15	0	DNC
Attached processes in partition p17				IPU		Board		
PID	Command	Time	User	ID	Clock	Temp	Temp	Power
907530	python	17s	u.zh108696	0	1500MHz	23.5 C	22.0 C	90.8 W

***Graphcore IPU systems** can be divided into **logical partitions**, each acting like an independent cluster of IPU's. Think of it like slicing a big GPU server into smaller virtual GPU "nodes" for different users or jobs.

Section III. Porting PyTorch Code to IPU



PopTorch

- PopTorch is a tool from Graphcore that enables PyTorch models to run on Graphcore's IPU hardware..
- You can also use a replication factor to apply data parallelism and use even more IPUs.

Training a model on IPU

Import the packages.

```
import torch
import poptorch
import torchvision
import torch.nn as nn
import matplotlib.pyplot as plt
from tqdm import tqdm
from sklearn.metrics import accuracy_score
```

Load the data

PopTorch offers an extension of `torch.utils.data.DataLoader` class with its `poptorch.DataLoader` class, specialized for the way the underlying PopART framework handles batching of data.

Build the Model

```
class ClassificationModel(nn.Module): def
    __init__(self):
        super().__init__()
        self.conv1 = nn.Conv2d(1, 5, 3)
        self.pool = nn.MaxPool2d(2, 2)
        self.conv2 = nn.Conv2d(5, 12, 5)
        self.norm = nn.GroupNorm(3, 12)
        self.fc1 = nn.Linear(972, 100)
        self.relu = nn.ReLU()
        self.fc2 = nn.Linear(100, 10)
        self.log_softmax =
nn.LogSoftmax(dim=1)
        self.loss = nn.NLLLoss()
```

```
def forward(self, x, labels=None): x =
self.pool(self.relu(self.conv1(x))) x =
self.norm(self.relu(self.conv2(x)))
    x = torch.flatten(x, start_dim=1) x =
    self.relu(self.fc1(x))
    x = self.log_softmax(self.fc2(x)) # The
    model is responsible for the
calculation of the loss when using an IPU. We do
it this way:
        if self.training:
            return x, self.loss(x, labels)
        return x

model = ClassificationModel()
model.train()
```

****In standard PyTorch, the loss is calculated outside the forward() method, while in Graphcore IPU (PopTorch), the loss is calculated **inside** the forward() method and returned. This enables PopTorch to handle loss computation and backpropagation on the IPU device automatically.**

Prepare training for IPUs

The **compilation and execution** on the IPU can be controlled using `poptorch.Options`. These options are used by PopTorch's wrappers such as `poptorch.DataLoader` and `poptorch.trainingModel`.

```
opts = poptorch.Options()
train_dataloader = poptorch.DataLoader(
    opts, train_dataset, batch_size=16, shuffle=True, num_workers=20
)
```

Train the model

```
optimizer = poptorch.optim.SGD(model.parameters(), lr=0.001, momentum=0.9)

poptorch_model = poptorch.trainingModel(model, options=opts,
optimizer=optimizer)

epochs = 30
for epoch in tqdm(range(epochs), desc="epochs"):
    total_loss = 0.0
    for data, labels in tqdm(train_dataloader, desc="batches", leave=False):
        output, loss = poptorch_model(data, labels)
        total_loss += loss

poptorch_model.detachFromDevice()

torch.save(model.state_dict(), "classifier.pth")
```

`**poptorch.trainingModel(model, ...)` wraps a standard PyTorch model to run training on Graphcore IPUs.

Evaluate the Model

```
model = model.eval() #set the model to evaluate mode

# Wrap model with poptorch.inferenceModel
poptorch_model_inf = poptorch.inferenceModel(model, options=opts)
test_dataloader = poptorch.DataLoader(opts, test_dataset, batch_size=32,
num_workers=10)          #Batches data for inference

predictions, labels = [], []
for data, label in test_dataloader:
    predictions += poptorch_model_inf(data).data.max(dim=1).indices
    labels += label

poptorch_model_inf.detachFromDevice()

print(f"Eval accuracy: {100 * accuracy_score(labels, predictions):.2f}%")
```


Hands-on Session

- **Activate the TF virtual environment**

```
cd $SCRATCH  
source poptorch_test/bin/activate
```

- **Change directory to PyTorch**

```
cd IPU-Training/PyTorch
```

- **Complete the #Todos** in the `fashion-mnist-pytorch-ipu-todo.py` file.

- **Run it in the poptorch_test virtual environment.**

```
pip install -r requirements.txt  
python fashion-mnist-pytorch-ipu-todo.py
```

- **After finishing the job, you can deactivate the virtual environment**

```
deactivate
```

To do Solution

- # Open the todo python file
`vim fashion-mnist-pytorch-ipu-todo.py`
- # Todo: Step 1 - import the PopTorch package.
`import poptorch`
- # Todo: Step 2 - add the loss computation
if self.training:
`return x, self.loss(x, labels)`
- # Todo: Step 3 - set up the poptorch.Options
`opts = poptorch.Options()`
- # Todo: Step 4 - create train_dataloader with poptorch.DataLoader.
`train_dataloader = poptorch.DataLoader(opts, train_dataset, batch_size=16, shuffle= True, num_workers=20)`
- # Todo: Step 5 - set up the optimizer with poptorch.optim.xxx# Write your code below
`optimizer = poptorch.optim.SGD(model.parameters(), lr=0.001, momentum=0.9)`

To do Solution

- #Todo: Step 6 - Use the `poptorch.trainingModel` wrapper for the `poptorch_model`. It takes `model`, `opts` and `optimizer`.
`poptorch_model = poptorch.trainingModel(model, options=opts, optimizer=optimizer)`
- # Todo: Step 7 - detach the model from the device to use the same IPU for training and inference
`poptorch_model.detachFromDevice()`
- # Todo: Step 8 - Use the `poptorch.inference Model` wrapper
`poptorch_model_inf = poptorch.inferenceModel(model, options=opts)`
- # Todo: Step 9 - create `test_dataloader` with `poptorch.Options`
`test_dataloader = poptorch.DataLoader(opts, test_dataset, batch_size=32, num_workers=10)`
- # Todo: Step 10 - detach the model from the device.
`poptorch_model_inf.detachFromDevice()`
- # Change the `print` function at the end of the script to:
`print("Eval accuracy: {:.2f}%".format(100 * accuracy_score(labels, predictions)))`

Vim - a text editor

1. ``vim your_file.py``

- This command opens the file named ``your_file.py`` in the Vim text editor. If the file does not exist, Vim will create a new file with that name.

2. ``I``

- This command is used in normal mode in Vim to switch to insert mode before the cursor. When you press ``I``, Vim will allow you to start typing text at the current cursor position.

3. ``ESC``

- The ``Esc`` key is used to exit from insert mode or any other mode and return to normal mode. It's a crucial key in Vim for transitioning between different modes.

4. ``:w`, `:wq`, `:q!``

- These are command-line commands you can use while in normal mode or command-line mode in Vim.
 - ``:w``: Writes (saves) the changes to the file without quitting Vim.
 - ``:wq``: Writes the changes to the file and quits Vim.
 - ``:q!``: Quits Vim without saving changes, discarding any modifications made.

Thank you !!